MILITARY SPECIFICATION

GRENADE, GP: M46
BODY ASSEMBLY, TYPE CA, FOR

This amendment forms a part of Military Specification MIL-G-48047B (AR), dated 21 December 1981, and is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

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4.4.2.2, Critical 1: Delete "4.5.4" from the Paragraph Reference/Inspection Method Column and substitute "4.5.5".

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4.4.2.3, Major 101: Delete "4.5.7" from the No. of Sample Units Column and substitute "4.5.6".

* Major 109: Delete the words "Bulge or".

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4.4.3.3a, Continuous Sampling: Delete in its entirety and substitute the following:

"a. Continuous Sampling - In order to establish an acceptable heat treatment process, take a group of thirty two (32) grenade bodies for each test from each heat treat furnace at the start of production or after any interruption in the process of one hundred (100) hours or more (a break in production shall be considered to be a period of time where no test samples are taken from material exiting the furnace). The groups of thirty two (32) shall be hardness tested in accordance with 4.5.2 (when in effect) and strength tested in accordance with 4.5.3 and 4.5.4 as applicable. Calculate the standard deviation(s) for each strength test using the formula in 6.6.1. If any individual value falls below the minimum load requirement, the process shall be corrected and additional groups of thirty two (32) grenade bodies taken. If all values are above the minimum load requirement, proceed with production.

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To qualify product during production, a continuous sampling plan shall apply which consists of randomly selecting a body every four (4) minutes for each test from each heat furnace and subjecting them to the tests of paragraph 4.5.3 and 4.5.4. Proceed in this manner until four (4) consecutive groups of four (4) minutes production have been collected into one lot representing sixteen (16) minutes production. The individual mean ($\bar{x}$) shall be calculated based on the values of the four test samples for each test. If any individual sample reading falls below the minimum load specified on the drawing or if the mean minus 3.12 times the standard deviation ($\bar{x} - 3.12s$) is below the minimum load requirement, then reject that lot and take corrective action. Rejected lots may be reheat treated. A suitable hold area shall be provided at each heat treat furnace.

The standard deviation shall be continually recalculated based on the values of the latest test samples, and individually numbered samples of twenty eight (28) grenade bodies previously tested (initial quantity obtained from above); and use of the formula in 6.6.1. The test statistics then form a running standard deviation based on thirty two (32) samples, including the most recent test samples as the twenty ninth thru thirty second samples and eliminating the first thru fourth samples. This process will continue for each additional group of bodies tested. One continuous data base shall be maintained for each grenade type per furnace.

Control charts shall be maintained as outlined in para 6.6.2. The contractor shall maintain these charts along with records of corrective action, available for government review, for a minimum of one year."

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4.4.4.1, Add new paragraph as follows:

"4.4.4.1 Certification of nondestructive test personnel. All personnel operating or calibratig non destructive test equipment shall be qualified and certified in accordance with MIL-STD-410."

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* 4.5.6, Delete in its entirety and substitute:

"4.5.6 Stud loose and pull test of clinch of stud on body assembly. As a minimum, one body assembly shall be sampled from each tool station of each machine from every 3800

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bodies assembled starting with the first unit produced at each production shift and whenever a new, repaired or adjusted crimping tool is put into use for production. If multiple tool stations are utilized and production from various tools is mixed, then a sample from every tool station must be taken for every 3800 bodies as outlined above. Each sampled body assembly shall be inspected manually first for loose studs and gap at junction of stud with top of cap. Any body assembly passing this test shall be placed in an approved fixture which will restrain the body. A tensile force, in an axial direction, shall then be applied to one of the studs and the stud must withstand the minimum force specified on the drawing. The test shall be repeated using the other stud on the body assembly. If any body assembly fails to comply with either requirement (loose stud or pull test), production from the tool which yielded the defective part since the last satisfactory test, shall be rejected. Where multiple tool stations are involved (when production from various tools is mixed) a failure of any sample to comply with either requirement will be cause to reject all 3800 bodies. Bodies from this test may be reworked to assemble new studs provided that the two stud holes are within the dimensional requirements of the drawing after the removal of the original studs and that the protective finish on the body has not been affected to expose the base metal during the rework process. Adequate hold areas shall be provided at each machine to accommodate parts produced between inspections. Accepted parts shall be released into the line for other inspections on the body assembly."

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5.1, 5.1.1, 5.2, 5.2.1 and 5.3: Delete in their entirety and substitute the following:

"5.1 Level B.

5.1.1 Preservation, packaging and marking. Grenades shall be packaged, packed and marked in accordance with dwgs. 9206166 and 9206167. Palletization is optional".

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* 6.6.1: Delete the $\overline{X}$ and N variable definitions and substitute the following: "$\overline{X}$ = average value of all individual observations, lbs (4.4.3.3b, Lot sampling); $\overline{X}$ = average value of the 4 individual observations, lbs.(4.4.3.3a. Continuous sampling). N = sample size (4.4.3.3b, Lot sampling); N = 4 = sample size (4.4.3.3a, Continuous sampling)"."

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6.6.2: Delete in its entirety and substitute the following:

"6.6.2 $\bar{x}$ & s control chart procedure. For both the longitudinal and transverse tests maintain $\bar{x}$ and s charts as follows.

6.6.2.1 Temporary control chart limits and center lines. After 25 subgroups have been plotted, calculate control chart center lines and control limits for the $\bar{x}$ chart and s chart as follows:

$\bar{x}$ chart center line = $\bar{\bar{x}}$ = average $\bar{x}$ of 25 subgroups of size n = 4
s chart center line = $\bar{s}$ = average s of 25 subgroups of size n = 4
$\bar{x}$ chart upper control limit (UCL$\bar{x}$) = $\bar{\bar{x}}$ + 1.628s
$\bar{x}$ chart lower control limit (LCL$\bar{x}$) = $\bar{\bar{x}}$ - 1.628s
s chart upper control limit (UCLs) = 2.27$\bar{s}$
s chart lower control limit (UCLs) = 0

6.6.2.2 Adjusted control chart limits. To monitor the heat treat process through statistical process control proceed as follows:

(a) Maintain the above control charts for the next 75 in control subgroups (a subgroup can be determined to be out of control by the criteria of 6.6.2.3). If an out of control condition arises follow the procedure outlined in 6.6.2.4.

(b) Recalculate $\bar{\bar{x}}$ and $\bar{s}$ using averages of 100 subgroups

(c) Using the new $\bar{\bar{x}}$ and $\bar{s}$ recalculate UCL$\bar{x}$, LCL$\bar{x}$ and UCLs

(d) Rechart the previous data with the new center lines and control limits

(e) Recalculate $\bar{\bar{x}}$, $\bar{s}$, UCL$\bar{x}$, LCL$\bar{x}$ and UCLs after removing the out of control subgroups from the calculations

(f) Proceed with charting the test results until 250 new in-control subgroups have been accumulated. Each time a subgroup is determined to be out of control follow the procedure outlined in 6.6.2.4.

(g) Calculate $\bar{\bar{x}}$ and $\bar{s}$ (based on 250 subgroups) for comparison purposes and make adjustments to the center lines and control limits based on the following conditions:

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(g.1) If $\bar{\bar{X}}$ has not decreased and $\bar{s}$ has not increased then recalculate $\bar{\bar{X}}$, $\bar{s}$, UCL$\bar{X}$, LCL$\bar{X}$ and UCLs (based on the new 250 subgroups).

(g.2) If $\bar{\bar{X}}$ has increased and $\bar{s}$ has increased then keep $\bar{s}$, and UCLs the same and recalculate $\bar{\bar{X}}$, UCL$\bar{X}$ and LCL$\bar{X}$ (based on the previous s value).

(g.3) If $\bar{\bar{X}}$ has not increased and $\bar{s}$ has increased or $\bar{\bar{x}}$ has decreased and $\bar{s}$ has not decreased then keep $\bar{\bar{x}}$, $\bar{s}$, UCL$\bar{x}$, LCL$\bar{X}$ and UCLs the same and follow the procedure of 6.6.2.4.

(g.4) If $\bar{\bar{X}}$ has decreased and $\bar{s}$ has decreased then keep $\bar{\bar{x}}$ the same and recalculate UCL$\bar{X}$, LCL$\bar{X}$, $\bar{s}$, and UCLs (based on the new 250 subgroups).

(h) Proceed with charting and repeat step (g) for every successive quantity of 250 new in-control subgroups accumulated.

6.6.2.3 Out of control criteria. The contractor shall consider the process out of control if any of the following occur.

(a) Any point falls outside of UCL or LCL
(b) Nine consecutive points are on the same side of a center line.
(c) Six consecutive points are either steadly increasing or decreasing.
(d) Fourteen points in a row are alternating up and down.

6.6.2.4 Reporting action. If any of the criteria of 6.6.2.3 are signaled, the contractor shall record in writing the believed cause of the out of control signal. The contractor shall also record the corrective action taken. If no corrective action is taken, the reason for not taking action shall be recorded. To facilitate investigations into causes of out of control conditions, the contractor shall annotate on the control chart whenever anything noteworthy happened."

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6.6.3: Delete in its entirety.

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